



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -1

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5. Bitwise operators (6)

- Bitwise operators perform on the binary representations of integers. **OR** Bitwise operators work on individual bits of integers. These are basically used for testing or shifting bits.

Operator	Description	Example
&	Binary AND Operator copies a bit to the result if it exists in both operands.	(A & B) will give 12, which is 0000 1100
	Binary OR Operator copies a bit if it exists in either operand.	(A B) will give 61, which is 0011 1101
^	Binary XOR Operator copies the bit if it is set in one operand but not both.	(A ^ B) will give 49, which is 0011 0001
~	Binary Ones Complement Operator is unary and has the effect of 'flipping' bits.	(~A) will give -61, which is 1100 0011 in 2's complement form. Result in decimal : ~A = -(A+1)
<<	Binary Left Shift Operator. The left operands value is moved left by the number of bits specified by the right operand.	A << 2 will give 240 which is 1111 0000
>>	Binary Right Shift Operator. The left operands value is moved right by the number of bits specified by the right operand.	A >> 2 will give 15 which is 0000 1111

```
int main()
{
    int a=5, b=3;
    printf("(a & b) = %d \n", (a&b)); //and
    printf("(a | b) = %d \n", (a|b)); // or
    printf("(a ^ b) = %d \n", (a^b)); // xor
    printf("(~a) = %d \n", (~a)); // bitwise negative OR Complement operator
    printf("(~b) = %d \n", (~b)); //bitwise negative
    printf("(a << 1) = %d \n", (a << 1)); // left shift operator
    printf("(a >> 2) = %d \n", (a >> 2)); // right shift operator
    getch(); // optional
}
```

```
C:\Users\hira\Desktop\hello.exe
(a & b) = 1
(a | b) = 7
(a ^ b) = 6
(~a) = -6
(~b) = -4
(a << 1) = 10
(a >> 2) = 1
```

Activate Window

Not Confuse

Bitwise	Logical
<code>a & b</code>	<code>a && b</code>
<code>a b</code>	<code>a b</code>
<code>a ^ b</code>	<code>a != b</code>
<code>~a</code>	<code>!a</code>

6. Assignment Operators (11)

- Assignment operators are used to assign a value (or) an expression (or) a value of a variable to another variable.
- Syntax : variable name=expression (or) value (or) variable

✓ Ex : `x=10;`
 `y=a+b;`
 `z=p`

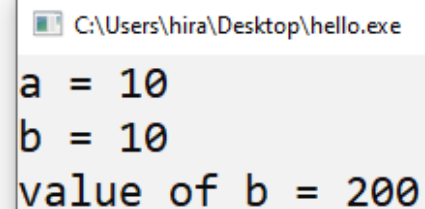
- `i += 2` (The operator `+=` is called an assignment operator.)

Operator	Example	Meaning
<code>+=</code>	<code>x += y</code>	<code>x=x+y</code>
<code>-=</code>	<code>x -= y</code>	<code>x=x-y</code>
<code>*=</code>	<code>x *= y</code>	<code>x=x*y</code>
<code>/=</code>	<code>x /= y</code>	<code>x=x/y</code>
<code>%=</code>	<code>x %= y</code>	<code>X=x%y</code>

<code>=</code>	Assignment operator
<code>+=</code> <code>-=</code>	Addition/subtraction assignment
<code>*=</code> <code>/=</code>	Multiplication/division assignment
<code>%=</code> <code>&=</code>	Modulus and bitwise assignment
<code>^=</code> <code> =</code>	Bitwise exclusive/inclusive OR assignment
<code><<=</code> <code>>>=</code>	

If `x *= y + 1` means $\rightarrow x = x * (y + 1)$ rather than `x = x * y + 1`

```
int main()
{
    int a = 10;
    int b = a;
    printf("a = %d \n", a);
    printf("b = %d \n", b);
    b*=a+b;
    printf("value of b = %d", b);
    getch(); // optional
    return 0;
}
```



```
C:\Users\hira\Desktop\hello.exe
a = 10
b = 10
value of b = 200
```

7. Ternary or Conditional Operator (1)

- In c, the ternary operator is a shorthand way to write simple if-else statement, It's the only operator in c that takes three operands.
- It is also called Conditional Operator.
- Syntax : Condition ? expression_if_true : expression_if_false;
- Example: int a=10,b=20;
 - Int max = (a>b) ? a : b ;

Explain this line

int max = (a>b) ? a : b ;



```
int max;  
if(a>b)  
{  
    max =a;  
}  
else  
{  
    max =b;  
}
```

```
#include<stdio.h> // printf() scanf()  
#include<conio.h> // getch()  
#include<stdbool.h> // bool  
int main()  
{  
    int a= 10, b=20;  
    int max_value = (a>b) ? a : b;  
    printf("Greatest Value = %d",max_value);  
    getch();  
    return 0;  
}
```

C:\Users\hira\Desktop\hello.exe

Greatest Value = 20